



A distributed intrusion detection framework for vehicular Ad Hoc networks via federated learning and Blockchain

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ABSTRACT

The emergence of connected vehicles via Vehicular Ad Hoc Networks (VANETs) has revolutionized transportation but has also brought forth challenges in security and privacy due to their open architecture. Early detection of intrusions within VANETs is paramount for ensuring safe communication. This research presents an intelligent distributed approach that leverages federated learning (FL) and blockchain for intrusion detection in VANETs. Through FL, various neural network models were implemented to distribute model training among vehicles, thus preserving privacy. Quantitative evaluation metrics demonstrate the effectiveness of the proposed framework. For example, compared to a traditionally trained Stochastic Gradient Descent (SGD) model, the Federated Trained Model achieved higher precision across various attack types, ranging from 68 % to 94 %, and consistently outperformed in terms of recall, with rates ranging from 57 % to 88 %. These results highlight FL's superiority in detecting intrusions, evidenced by gains in accuracy, recall, and precision. Integration of FL with blockchain further strengthened security and privacy protection, ensuring data integrity during collaborative FL training across decentralized nodes. This novel framework addresses VANET vulnerabilities by facilitating privacy-preserving, collaborative anomaly monitoring in a trustworthy manner. Evaluations validate the performance advantages of FL for intrusion identification, supporting wider adoption of vehicular technologies. The study underscores the potential of combining FL and blockchain to enable robust, cooperative abnormality recognition crucial for maintaining reliability, safety, and trust in VANET operations.

1. Introduction

The rapid evolution of technology has revolutionized transportation systems worldwide, with Vehicular Ad Hoc Networks (VANETs) emerging as integral components of intelligent transportation infrastructures [1,2]. By enabling seamless communication between vehicles and infrastructure, VANETs promise safer and more efficient transportation. However, the inherent openness and dynamism of VANETs introduce significant security and privacy challenges, underscoring the critical importance of intrusion detection mechanisms [3].

VANETs play a pivotal role in the advancement of smart cities, fostering collaboration between academia and industry to enhance various aspects of urban life, including security, safety, and quality of life [4]. As such, understanding the intricacies of VANETs, including

intelligent transport systems (ITS), smart vehicles, and vehicular network architectures, is paramount for addressing emerging challenges and harnessing the full potential of these networks [5]. In response to the pressing need for intrusion detection in VANETs, this paper proposes an intelligent approach that combines federated learning (FL) and blockchain technology. FL has garnered attention as a collaborative learning paradigm that facilitates decentralized model training without centralized data aggregation, thereby ensuring privacy preservation and data security [6]. However, the resource-constrained nature of VANET nodes necessitates a deeper exploration of model complexity and optimization techniques to ensure efficient computation and real-time responsiveness. Our research endeavors to address these challenges by delving into the intricacies of neural network models tailored for intrusion detection in VANETs. We aim to unravel strategies for

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optimizing model complexity and computational efficiency, thereby enhancing the practical applicability of our proposed framework. The use of neural network models for intrusion detection in VANETs introduces some complexity in terms of designing the models and optimizing them to ensure efficient calculations. The complexity of neural network models can be evaluated from different perspectives. First, it is crucial to consider the number of layers and depth of the network. Deep neural networks, with many intermediate layers, can capture complex and abstract features. However, this can also increase the complexity of the calculations. Therefore, it is essential to strike a balance between network depth and the processing capacity of VANET nodes. The size of the layers and the number of neurons in each layer can contribute to the complexity of the model. Layers with many neurons can facilitate the learning of richer representations, but this also implies an increase in memory and calculation requirements. It is therefore important to find a compromise between the complexity of the model and the limited resources of the VANET nodes. In addition to the inherent complexity of the model, it is imperative to delve into optimization techniques that ensure efficient computation. Model compression is one such technique that focuses on reducing the model's size by eliminating redundancies and adopting more compact representations. This can be accomplished through methods like weight compression, neuron pruning, and weight quantification. Parallel computing presents another optimization approach. Given the capacity of VANET nodes to leverage parallel computing capabilities, employing techniques such as distributed processing or hardware acceleration can substantially enhance the efficiency of computations. Additionally, the integration of blockchain technology into the FL framework introduces further complexities and optimization considerations, particularly concerning the dynamic and resource-constrained nature of VANETs. The proposed methodology not only enhances the security and privacy of vehicular communication but also contributes to the advancement of intrusion detection techniques in VANETs. By leveraging the VeReMi dataset and conducting comprehensive experimental evaluations, we demonstrate the feasibility and effectiveness of our approach in accurately detecting intrusions while preserving privacy and security.

Our paper offers valuable insights into the challenges and potential solutions for enhancing intrusion detection mechanisms in VANETs. By combining the strengths of FL and blockchain, our research contributes to the resilience and security of future intelligent transportation systems. The subsequent sections delve into our methodology, experimental results, and conclusions, offering valuable insights for researchers, practitioners, and policymakers in the field.

2. Related work

ML-based VANET systems rely on sensitive information for decision-making, but privacy mechanisms are employed to protect this data. However, resource-intensive On-Board Units (OBUs) make traditional ML and misuse-based learning approaches impractical for vehicle networks. Therefore, a lightweight protocol for secure information transfer is essential in VANET [6]. From a resource perspective, high-performance computational resources are necessary to train DL-based models. DL has gained significant attention and is widely used in various domains of VANET to improve the performance of tasks such as anomaly detection in vehicular communication, network traffic classification, and attack detection [7]. DL models automatically analyze and extract useful features from the provided data. In the context of VANET, DL-based models achieve high performance using raw traffic inputs and minimal resources. This makes DL a suitable technique for application in vehicular networks. Table 1 compares recent and related work on VANETs using ML and DL methods.

Nonetheless, DL and ML algorithms in VANET face limitations related to the availability of labeled training data, computational complexity, security vulnerabilities, and interpretability. To overcome these challenges, federated learning (FL) offers a promising solution. FL

Table 1

Comparison of recent research in VANETs using ML and DL methods.

Reference	Dataset	Methodology
Steven et al. [8]	VeReMi	The author introduced a framework that incorporates plausibility checks as a feature vector for machine learning models, enhancing the detection and classification of misbehavior. By employing KNN and SVM, the results indicate a significant improvement of over 20 % in the overall detection precision of the plausibility checks, while maintaining a recall within 5 %.
Zeng, Yi, et al. [6]	NS-3 VANET simulated dataset and ISCX 2012 IDS dataset	The authors proposed a DL-based system to automatically detect malware traffic for OBUs. Unlike previous techniques, this method only requires raw traffic, without the need for human-extracted private information attributes.
Shu, J. et al. [9]	KDD99 dataset	The authors proposed the installation of a distributed SDN controller on each base station to establish a cooperative intrusion detection system using distributed SDN.
Hsu, H. et al. [10]	VeReMi extension	The authors presented an integrated algorithm for misbehavior detection systems (MDS). It combines convolutional neural network (CNN) and long short-term memory (LSTM) models to reconstruct location information. Additionally, a support vector machine (SVM) is used for binary classification to determine vehicle compromise.
Yu, Y. et al. [11]	Time series dataset	The authors introduced an intrusion detection system (IDS) based on time series classification and deep learning to improve the detection of false emergency messages. They utilized long short-term memory (LSTM) to construct and train a classifier that accurately identifies the authenticity of emergency messages.
Hossain, S. et al. [12]	VeReMi Extension	The authors proposed a lightweight intrusion detection scheme for 5G-V2X sliced networks. It utilizes DL and Knowledge Distillation (KD) to train cloud-based models and transfer knowledge to lightweight models on CAVs. This approach balances detection accuracy and security overhead, reducing computation and memory complexity while maintaining performance comparable to heavier DL-based IDSs.

allows collaborative model training while preserving privacy by not sharing raw data. It leverages the distributed nature of the network, enabling vehicles to contribute to training while keeping data local. FL improves resource efficiency, reduces data transmission needs, and enhances model generalization through knowledge aggregation from multiple vehicles in real-world VANET scenarios.

Federated learning (FL) has been recognized as a solution to handle centralized data challenges in critical networks like VANETs. It serves as a distributed machine learning model that leverages the benefits of AI-based computation applications in devices such as smart cars, while also ensuring end-user privacy. The FL process involves training a global ML/DL model and distributing updated parameters to end devices via a centralized server. Each device then trains its local ML/DL model on its data, utilizing the guidance of the trained model. After training the local models, clients send their parameters back to the centralized server for

global training. This iterative process continues until completion, accommodating the demands and precision requirements of the centralized servers. FL finds utility in various applications, including e-commerce and e-healthcare, as depicted in Fig. 1 [13]. In summary, FL's deployment in VANETs and other contexts allows for distributed learning, privacy preservation, and efficient utilization of local data for improved machine learning outcomes.

Federated learning, a decentralized ML approach, can be applied to VANET environment to improve various aspects of the network. VANETs consist of vehicles equipped with wireless communication capabilities, allowing them to exchange information with each other and with roadside infrastructure. Here are some potential use cases and benefits of using federated learning in a VANET environment:

Traffic Prediction: FL can be used to predict traffic patterns and congestion in a VANET.

Collaborative Collision Avoidance: Vehicles in a VANET can use FL to collaboratively learn collision avoidance strategies.

Anomaly Detection: FL enables the detection of anomalies in a VANET, such as the identification of malicious or malfunctioning vehicles.

Energy Optimization: In a VANET, FL can optimize energy consumption by developing models that consider vehicle-specific parameters like battery level and power usage.

Cooperative Sensing: FL can facilitate cooperative sensing in a VANET, where vehicles collaboratively gather and share information about road conditions, weather, or other relevant environmental factors.

Privacy-Preserving Learning: Privacy is a significant concern in VANETs due to the sensitive nature of vehicle data. FL can be employed to address this issue by training models locally on individual vehicles, without transmitting raw data to a central server.

Many previous works that aim to improve the detection of intrusions, traffic Prediction, energy consumption and Privacy-Preserving Learning are illustrated in the Table 2.

Federated learning poses unique challenges when applied to VANETs. The dynamic and decentralized nature of VANETs, coupled with limited network connectivity, varying data distributions, and highly mobile nodes, present significant hurdles for implementing federated learning in this context. Issues such as network latency, communication overhead, synchronization, and ensuring privacy in a distributed environment further compound the complexity of deploying federated learning in VANETs. Consequently, addressing these challenges and developing efficient federated learning techniques specifically tailored for VANETs remain active areas of research and development.

Furthermore, Blockchain garnered significant attention from both academia and industry due to its notable benefits in addressing data transfer limitations among diverse devices within the Internet of Vehicles (IOV), while ensuring privacy and security [23]. Originally developed for banking and digital currencies, blockchain has found applications in various areas. Intelligent Transportation Systems (ITS) have alleviated urban transportation problems, thanks to wireless

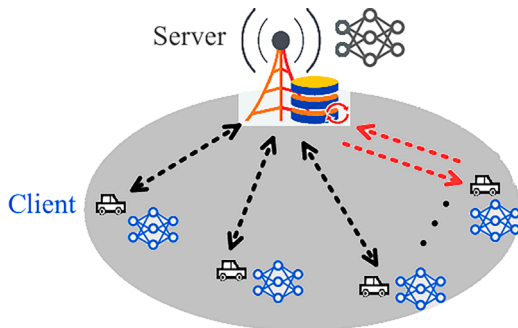


Fig. 1. Usage of federated learning in a VANET environment [14].

Table 2

Previous works in the VANET environment that aim to improve the detection of intrusions, traffic Prediction, energy consumption, and Privacy-Preserving Learning.

Use cases	Contribution
Traffic Prediction	Wireless traffic prediction based on FL (FedDA). [15] Vehicular speed and traffic flow prediction in FL way based on LSTM (FedTSE). [16]
Anomaly Detection	Intrusion detection in smart cities using distributed federated learning (FL) of neural networks [17]. Intrusion Detection Systems (IDSs) in the IoV using FL [18]
Energy Optimization	Unmanned Aerial Vehicles (UAVs) energy efficiency and Unmanned Ground Vehicles (UGVs) connectivity using FL [19] Comparison of Federated learning framework (FedAvg) and HFL in terms of energy consumption and communication delay and accuracy[20].
Privacy-Preservation	Anonymous identity-based privacy preserving scheme to protect the identity privacy of vehicles with Zero-Knowledge Proof (ZKP) using FL. [21] Decentralized federated learning (FL) scheme that incorporates a quality-based aggregation method, extended dynamic contribution broadcast encryption (DConBE), and local differential privacy. [22]

sensors in vehicles enabling Vehicular Ad-hoc Networks (VANETs) [24]. VANETs enhance network traffic flow, providing timely information and intelligent transportation services. VANETs facilitate self-organized data transmission for on-road vehicles, supporting applications such as safety warnings and in-vehicle driving assistance.

The European Parliament has adopted Dedicated Short-Range Communications (DSRCs) instead of cellular vehicular-to-everything (C-V2X) technology [25]. Blockchain is a novel technology that facilitates peer-to-peer transactions and influences distributed ledgers without a central authority. Its potential value in enhancing data credibility and accountability has been recognized in various VANET applications. Vehicles, operating autonomously, require increased trust in data exchange, particularly in intelligent transportation scenarios like smart contracts [26].

In blockchain technology, nodes validate new blocks to establish trust within the system. This validation process, known as Proof-of-Work (PoW), ensures the validity of certain transactions in the network. Full Nodes (FNs) play a crucial role in verifying newly created blocks by broadcasting them to all FNs in the network. This process affects blockchain performance by considering factors such as total broadcast delay and the number of nodes. Security, consensus credibility, and energy consumption are vital considerations in blockchain networks [27].

To ensure the identity and privacy of vehicles and their locations in VANETs, an architecture based on the IoT chain approach and private blockchain was constructed [28]. The primary objective of VANETs is to enhance credibility and security in data exchange. In this regard, a three-layer framework was proposed to improve data security [29]. To address privacy concerns, an anonymous reputation system based on blockchain was suggested to dissociate public keys from real identities [30]. Blockchain was also employed to ensure secure data communication among flying drones in a 5 G network [31]. Additionally, a consortium blockchain technique was studied for secure storage systems and data sharing in VANETs [32]. Another research work utilized smart contracts and consortium blockchains to achieve secure data sharing and storage within VANETs [33]. Addressing privacy and security concerns, the study assumed edge computing servers, including RSUs, faced significant challenges and were not entirely trusted. Furthermore, blockchain technology was utilized to enhance edge computing in electric vehicles [34]. Context-aware applications were recognized based on the energy and information interactions between vehicles, utilizing metrics such as energy involvement and data involvement frequency to achieve Proof-of-Work (PoW).

However, traditional ML and DL approaches are inadequate for

intrusion detection in VANETs due to the dynamic and complex nature of these networks. They struggle to provide adaptive and real-time detection mechanisms, and the heterogeneity of vehicular data poses additional challenges. In contrast, FL offers a promising solution. FL enables vehicles to collaboratively train a model while preserving privacy by not sharing raw data. This approach harnesses the collective intelligence of vehicles, enhancing intrusion detection in VANETs and improving scalability, privacy, and computational efficiency.

Similarly, Blockchain technology enhances security and trust in VANETs through its decentralized and tamper-resistant nature. It enables transparent and secure data storage and verification. Integration of blockchain with FL ensures secure and verifiable aggregation of model updates. The immutable blockchain ensures the integrity and authenticity of intrusion detection data, enhancing overall system reliability. Integrating FL and blockchain is crucial to harnessing the strengths of both approaches. By combining FL's privacy-preserving and collaborative learning with blockchain's tamper resistant data management, a robust and efficient solution for intrusion detection in VANETs is achieved. This integrated method overcomes limitations of traditional ML and DL methods, offering improved adaptability, privacy, security, and reliability for intrusion detection in VANETs.

Vehicular networks (VN) handle extensive data transmission, storage, and real-time computations for entertainment and safety purposes. It enables communication between vehicles and infrastructure (V2I), vehicles themselves (V2V), and everything else (V2X). V2I relies on VANET and RSU for reliable information. V2V is used during rush hours or emergencies without infrastructure. AI, ML, and fuzzy logic are employed for vehicle direction, speed, and propagation loss, but these methods require significant processing power, which is not feasible in a fully distributed architecture. FL is the ideal choice for such problems due to its distributed machine learning approach. Mobile devices collect data, train local models, and share them with an aggregator to generate a global model. Mobile devices continue training the global model until the desired goal is reached. Integrating FL and blockchain addresses challenges in centralized systems, such as resource limitations and is suitable for autonomous vehicle applications. This decentralized solution optimizes resource utilization and balances workloads compared to centralized networks. Additionally, using cryptography in VANET ensures data consistency, secure storage, and transmission within the network. Blockchain tracks connected vehicles and verifies their identity, enabling data authentication and authorization. It ensures secure data communication among vehicles, validates transactions, and saves locally trained data for each vehicle [13].

The disclosure of valuable data through model parameters during data transmission has prompted researchers and developers to enhance the security of VANET. They have adopted well-known security and privacy defense methods, integrating FL and BC technologies, to safeguard against reverse engineering and protect sensitive information. Table 3, illustrates some previous works on this field.

In this work, we are specifically interested in integrating federated learning and blockchain in VANETs to resolve the detection of intrusions using the VEREMI dataset. By combining the collaborative model training capabilities of federated learning and the decentralized and tamper-resistant features of blockchain technology, we aim to enhance the security and privacy of VANETs. The VEREMI dataset serves as a valuable resource for evaluating the effectiveness of our proposed approach in accurately and efficiently detecting intrusions in VANETs.

The subsequent section will present our proposed method for integrating FL, blockchain, and neural networks for detecting intrusions and improving the security of VANET networks using the Veremi dataset, along with the results obtained.

3. Proposed model of intrusion detection

We develop an integrated approach that addresses the intrusion detection challenges of VANET. This section provides an overview of our

Table 3

Related works on integrating federated learning and blockchain in VANET.

Problems	Related works description
Security	In 2019, Qiu, Junfei, et al. [35], introduced an innovative secure spectrum-sharing technique for VANET cellular networks using blockchain. In 2019, the authors in [36], proposed "FLchain," a blockchain-based model to enhance the security of federated learning (FL). In 2022, the authors in [37], proposed FBTM, combining federated learning and blockchain, for trust management in IoV. Simulation results show its effectiveness in enhancing IoV security.
Privacy Preservation	In 2020, the authors in [38], designed an autonomous transport system with privacy preservation and efficient vehicle-to-vehicle communication. By applying a blockchain consensus scheme, central coordination was eliminated. In 2021, Ghimire and Rawat. in [39], presented a theoretical framework for blockchain-empowered FL in IoV was proposed, prioritizing privacy, verifiability, and security in ITS services. Blockchain facilitates model parameter exchange, while miner vehicles ensure secure federation aggregation. In 2022, Moulahi, Tarek, et al. in [40], utilized classification techniques to detect cyber threats in VANET and ITS at the vehicle level. Through blockchain and an aggregation strategy, models are uploaded, updated, and returned to vehicles.
Energy Optimization	Unmanned Aerial Vehicles (UAVs) energy efficiency and Unmanned Ground Vehicles (UGVs) connectivity using FL. [19] Comparison of Federated learning framework (FedAvg) and HFL in terms of energy consumption and communication delay and accuracy. [20].
Privacy-Preservation	Anonymous identity-based privacy preserving scheme to protect the identity privacy of vehicles with Zero-Knowledge Proof (ZKP) using FL. [21] Decentralized federated learning (FL) scheme that incorporates a quality-based aggregation method, extended dynamic contribution broadcast encryption (DConBE), and local differential privacy. [22]

proposed solution and we outline the high level of the proposed method. Then, we describe the implementation details of blockchain integration.

3.1. Overview of veremi dataset

VANET is actively researched in modern vehicular networks. Previous studies focused on detecting misbehavior, intrusion, and security attacks. However, most research used private datasets and tools, making comparisons difficult. To address this issue, Heijden et al. [41], introduced VeReMi (Vehicular Reference Misbehavior Dataset), which is the first public and extensible dataset for VANET research. VeReMi serves as a baseline for detecting position falsification attacks and consists of 225 individual simulations. It includes five different types of attackers, three attacker densities, three traffic densities, and five repetitions of each parameter set with random seeds. The dataset is built on the Luxembourg SUMO traffic scenario (LuST) using VEINS and OMNET++ tools. To differentiate between legitimate vehicles and attacker vehicles, an attacker-type value is assigned. Legitimate vehicles have an attacker type of 0, while different attacks are represented by values 1, 2, 4, 8, and 16 as shown in Table 4. The VeReMi dataset covers various traffic scenarios, including highways, cities, and streets. In the present research, the VeReMi dataset is employed with different models using federated learning and SGD approaches to classify misbehavior for the five different types of attacks.

The VeReMi dataset is a valuable resource for gaining insights into vehicle behavior, enabling the identification of anomalies and potential attacks in VANETs. It offers a wide range of features, including velocity components, displacements, and other attributes that facilitate the construction of accurate intrusion detection models. By leveraging this dataset, the security of connected vehicle communications can be enhanced, instilling greater confidence in the system's integrity.

A thorough scalability analysis was conducted to assess the

Table 4
Comparison of recent research in VANETs using ML and DL methods.

Attacker	Type	Attack Name Description
1	Constant	The vehicle initially transmits its fixed position. To enhance privacy and prevent tracking, an offset is added to the vehicle's actual position before transmission. This offset introduces a level of uncertainty regarding the vehicle's true location..
2	Constant Offset	To further protect privacy, the vehicle periodically transmits a random position within a simulation area. This random position helps to obfuscate the vehicle's true whereabouts and adds an element of unpredictability.
4	Random	The vehicle may transmit a random position from a preconfigured rectangular area surrounding its actual location. This approach restricts the transmitted positions to a specific region, ensuring that the vehicle's true position remains concealed while still providing some contextual information.
8	Random Offset	There is a potential threat of an attacker compromising the system. Initially, the attacker behaves normally, transmitting positions consistent with their actual location. This behavior helps the attacker blend in and avoid suspicion. But after some time, the attacker changes their strategy and starts repeatedly transmitting their current position. This repetitive transmission can be a deliberate attempt to disrupt the system, compromise privacy, or potentially launch an attack.
16	Eventual Stop	To mitigate such attacks, robust security measures should be in place, including authentication mechanisms, encryption of transmitted data, and anomaly detection algorithms. These measures can help identify suspicious behavior and prevent malicious actors from compromising the system's integrity and privacy.

performance of the proposed approach in a large-scale VANET environment, utilizing the VeReMi dataset. VeReMi is a valuable dataset that provides comprehensive insights into the behavior of vehicles in VANETs. It encompasses various attributes, including velocity components, total displacement, and constant offset checking, which play a crucial role in constructing precise intrusion detection models and enhancing communication security among connected vehicles. The scalability analysis focused on evaluating how the proposed approach handles varying network sizes and densities, adapts to dynamic mobility patterns, and optimally utilizes available resources. By leveraging the VeReMi dataset, which offers a substantial number of instances and a comprehensive representation of VANET traffic, the analysis provided detailed insights into the protocol's scalability and suitability for real-world VANET deployments. The performance metrics considered during the analysis included packet delivery ratio, end-to-end delay, network throughput, and the ability to detect and mitigate attacks. Evaluating the protocol's performance under increasing network sizes showcased its ability to maintain satisfactory performance levels as the number of vehicles in the network grew. Additionally, the analysis assessed the protocol's efficiency in handling varying network densities, ensuring effective communication even in highly dense scenarios. Furthermore, the analysis took into account the impact of mobility patterns on the protocol's performance. By considering realistic mobility patterns present in the VeReMi dataset, the study evaluated how the protocol adapted to dynamic changes in network topologies and vehicle trajectories. This assessment verified the protocol's ability to handle dynamic movements and maintain reliable communication in large-scale VANET environments.

Table 4 presents a subset of the VeReMi dataset, comprising attributes such as x-velocity, y-velocity, constant-offset-check, total displacement (total movement), and attack type (ranging from 0 to 5 attack types). With a total of 105,320 instances and 5 columns, this dataset provides a comprehensive representation of VANET network

traffic. Utilizing this dataset enables conducting an in-depth analysis of network traffic scalability, taking into account the dynamic nature of VANET networks. By utilizing the VeReMi dataset, our approach demonstrates its effectiveness in handling VANET networks of varying sizes and densities. This choice of dataset allows us to showcase the scalability of our approach and its ability to adapt to the dynamic nature of VANET networks (Table 5).

3.2. NNFLB-NET: Proposed model

Our model aims to enhance intrusion detection capabilities in VANET by implementing an improved research strategy. We seek to address the security challenges associated with VANET by focusing on two main components: federated learning and Blockchain technologies.

3.2.1. Integrating federated learning with blockchain

Integrating federated learning with blockchain can provide additional mechanisms to ensure data integrity and privacy in VANET networks. Firstly, data encryption is a crucial step. Before sharing data with federated members, it can be encrypted using cryptographic techniques such as symmetric or asymmetric encryption. This ensures that the data remains secure and confidential during transmission and storage. Next, secure data aggregation plays a vital role in federated learning. Blockchain can be used to securely record and verify the aggregation process. Each client's model update can be hashed and recorded on the blockchain, making the entire aggregation process tamper-proof and providing an immutable record of the aggregation. A consensus mechanism is also important to ensure data integrity. By integrating federated learning with blockchain, a consensus mechanism like proof-of-work or proof-of-stake can be used to ensure that only valid and authenticated model updates are included in the aggregation process. This prevents malicious clients from manipulating the learning process.

Smart contracts are crucial in privacy-preserving aggregation. Smart contracts can incorporate differential privacy techniques to add noise or perturbations to the aggregated model updates. This protects the privacy of individual clients' data while still allowing useful information to be shared and utilized for training. Blockchain-based access control mechanisms can be employed to manage data access and permissions. Clients can define smart contracts that specify who can access their data and under what conditions. This allows clients to maintain ownership and control over their data while participating in the federated learning process. Auditability and transparency can be achieved through the use of blockchain. The blockchain provides transparency and auditability of transactions, allowing participants to track and verify the integrity of model updates and aggregation results. Clients and other network participants can audit the blockchain to ensure that the aggregation process follows the defined protocols and that the resulting global model is trustworthy.

Incentive mechanisms can be implemented through smart contracts. Clients can be rewarded or incentivized for their participation in the federated learning process, such as contributing high-quality data or model updates. This encourages active participation and enhances the overall security and quality of the learning process.

By integrating these technical mechanisms, federated learning with blockchain can provide enhanced data integrity, privacy, transparency, and incentive mechanisms in VANET networks, addressing the security challenges associated with vehicular communication.

In federated learning, model aggregation refers to the process of combining the model updates from multiple clients participating in the learning process to create a global model. Each client trains a local model using its own data and shares the model updates with a central server or coordinator. The server then aggregates these updates to create a refined global model that incorporates knowledge from all the clients while preserving data privacy.

Blockchain consensus mechanisms play a crucial role in ensuring the integrity and trustworthiness of the model aggregation process.

Table 5
Overview of VeReMi dataset.

ID	Velocity_x	Velocity_y	Constant_Offset_Check	Total_Displacement	Attack Type
0	-3.362600	29.137000	29.330000	0.000	5
1	10.013000	0.621020	10.033000	0.000	1
2	0.283390	0.018390	0.283990	217.200	0
3	-0.056932	0.214690	0.222120	68.672	2
4	0.276000	-0.191970	0.336190	69.873	4
5	0.119970	1.070000	1.076700	183.390	3

Consensus mechanisms provide a way for network participants to agree on the validity of model updates and prevent malicious actors from manipulating the learning process. One commonly used consensus mechanism is proof-of-work (PoW), where network participants compete to solve complex mathematical puzzles. The first participant to solve the puzzle adds the next block of transactions to the blockchain, and this block is considered valid by the network. PoW ensures that the majority of participants in the network agree on the validity of the model updates, making it difficult for a malicious actor to tamper with the aggregation process without expending significant computational resources.

Another consensus mechanism is proof-of-stake (PoS), where participants are chosen to create new blocks based on their stake or ownership in the network. In PoS, the probability of being selected to create a new block is proportional to the participant's stake. This mechanism provides a more energy-efficient alternative to PoW while still maintaining the security and integrity of the blockchain.

By integrating federated learning with blockchain consensus mechanisms like PoW or PoS, the model aggregation process becomes more robust and resistant to manipulation. The consensus mechanism ensures that only valid and authenticated model updates are included in the aggregation, enhancing the overall security and trustworthiness of the federated learning system in VANET networks.

3.2.2. Steps of the NNFLB-NET model

We employ a step-by-step model, as illustrated in Fig. 2, to guide our work process. The initial step involves inputting the VeReMi dataset. Subsequently, in the second step, we create federated members known as clients. For our study, we have chosen 10 clients. The data of these clients is then prepared for the training phase. Moving forward, we proceed to implement neural network models, including, MLP, LSTM, GRU, MANN, and CW-RNN. The subsequent crucial step involves integrating federated learning technology, utilizing the aggregation method called FedAVG.

This methodology involves the computation of a weighted average of the local model weights, resulting in a global model that accurately represents all participating clients. Once obtained, this global model undergoes training, testing, and evaluation processes to detect potential attacks. In the final step, we seamlessly integrate blockchain technology into the system through the implementation of smart contracts.

By implementing blockchain-based smart contracts in the context of federated learning on VANET networks, we achieve a transparent and secure automation of the incentive and reward process. Here are further details on how they operate:

Smart contract creation: Initially, a smart contract is developed and deployed on the blockchain. This smart contract incorporates programmed logic to effectively manage incentives and rewards for vehicle participation in federated learning.

Recording contributions: Each participating vehicle submits its contributions to the smart contract. These contributions encompass shared data, provided computing power, or any other specific contribution required by the contract. Contributions are recorded as transactions on the blockchain.

Reward calculation: The smart contract utilizes the recorded contributions to calculate corresponding rewards. Calculation rules can either be predefined within the contract or flexibly programmed to adapt to specific system requirements. For instance, rewards can be based on the amount of data provided by each vehicle or on an assessment of contribution quality.

Reward distribution: Once rewards are calculated, the smart contract distributes them to the participating vehicles. This distribution process generally involves transferring digital assets, such as tokens or cryptocurrencies, to the vehicle addresses on the blockchain. The distribution occurs transparently and securely, without requiring additional human intervention.

Security and immutability: Blockchain-based smart contracts provide a high level of security and immutability. The predefined rules and conditions within the contract are automatically executed and cannot be modified once deployed on the blockchain. This ensures fair and transparent reward distribution, eliminating the risk of manipulation or fraud.

Transparency and traceability: All transactions and operations are recorded on the blockchain, resulting in a transparent and traceable incentive and reward process. Participants can verify their contributions, received rewards, and transactions at any time, fostering trust in the system.

By integrating smart contracts into federated learning on VANET networks, we establish an automated, secure, and transparent environment that encourages participation and rewards vehicle contributions. This eliminates the need for trusted intermediaries and ensures the

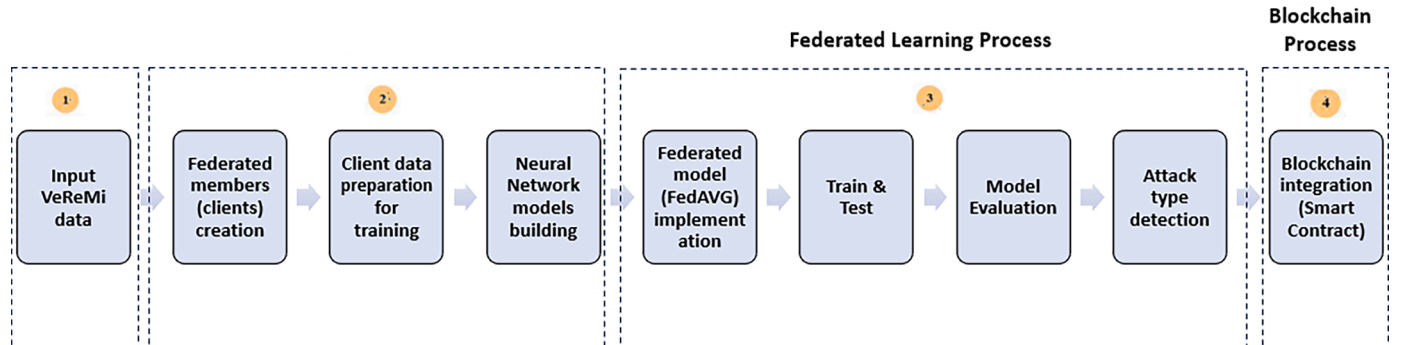


Fig. 2. Steps of NNFLB-NET model.

fairness, efficiency, and reliability of the incentive and reward process. This decentralized trust model promotes a secure and tamper-proof environment, enhancing the overall reliability and transparency of the system.

3.2.3. Proposed architecture

Our proposed architecture, illustrated in Fig. 3, aligns with our research objectives. The primary focus of our work is to integrate federated learning and blockchain technologies while utilizing various neural network models to predict intrusions in VANET based on the input VeReMi dataset.

The proposed model has three main stages: neural network models, the Federated Averaging model, the integration of the federated learning model with blockchain, and the integration of federated learning with homomorphic encryption. A detailed discussion of these four stages is as follows.

We used different neural networks including, MPL, LSTM, GRU, MANN, and CW-RNN to address the research objectives. Specifically, we utilized a federated learning approach and a standard Stochastic Gradient Descent (SGD) approach for training these neural network models.

The specified MLP model consists of three layers with ReLU and softmax activation functions. The model complexity is relatively low due to the limited number of layers and units. To optimize the MLP model, several techniques can be considered. These include appropriate weight initialization methods like Xavier or He initialization, exploring alternative activation functions such as Leaky ReLU or ELU, applying regularization techniques like L1 or L2 regularization to prevent overfitting, utilizing dropout to improve generalization, employing batch normalization for improved training stability, experimenting with different optimizers like Adam or RMSprop, and conducting a systematic hyper-parameter tuning to optimize model performance. By applying these optimization techniques, the MLP model can be enhanced in terms of training efficiency and generalization capabilities. However, it is important to note that the effectiveness of these techniques may vary

depending on the specific dataset and problem at hand, and iterative experimentation and adjustments may be required to achieve the best results.

The LSTM model specified consists of three layers, with the first two layers being LSTM layers and the final layer being a Dense layer with a softmax activation function and six units. The model complexity is influenced by the number of layers, the number of LSTM units in each layer, and the connections between them. By having LSTM layers, the model is capable of capturing long-term dependencies and sequential patterns in the data. To optimize the LSTM model, various techniques can be employed. These include careful weight initialization, selecting appropriate activation functions, implementing regularization methods such as L1 or L2 regularization, utilizing dropout to prevent overfitting, applying gradient clipping to address exploding gradients, experimenting with learning rate scheduling, monitoring validation loss for early stopping, and optimizing memory usage in LSTM cells. These optimization techniques aim to improve the efficiency of training, enhance generalization capabilities, and mitigate common challenges associated with LSTM models. It is important to note that the effectiveness of these techniques may depend on the specific dataset and problem, and thorough experimentation and evaluation are essential to determine the optimal configuration for achieving the desired model performance.

The GRU model specified comprises two layers, with the first layer being a GRU layer consisting of 16 units, and the second layer being a Dense layer with six units and a softmax activation function. The model complexity is influenced by the number of layers, the number of units in the GRU layer, and the connections between them. GRU (Gated Recurrent Unit) is a type of recurrent neural network (RNN) that can effectively capture sequential dependencies in the data. To optimize the GRU model, various techniques can be applied. These include appropriate weight initialization methods, selecting suitable activation functions, incorporating regularization techniques like L1 or L2 regularization to prevent overfitting, employing dropout for regularization and improved generalization, implementing gradient clipping to address exploding gradients, exploring learning rate scheduling strategies, monitoring

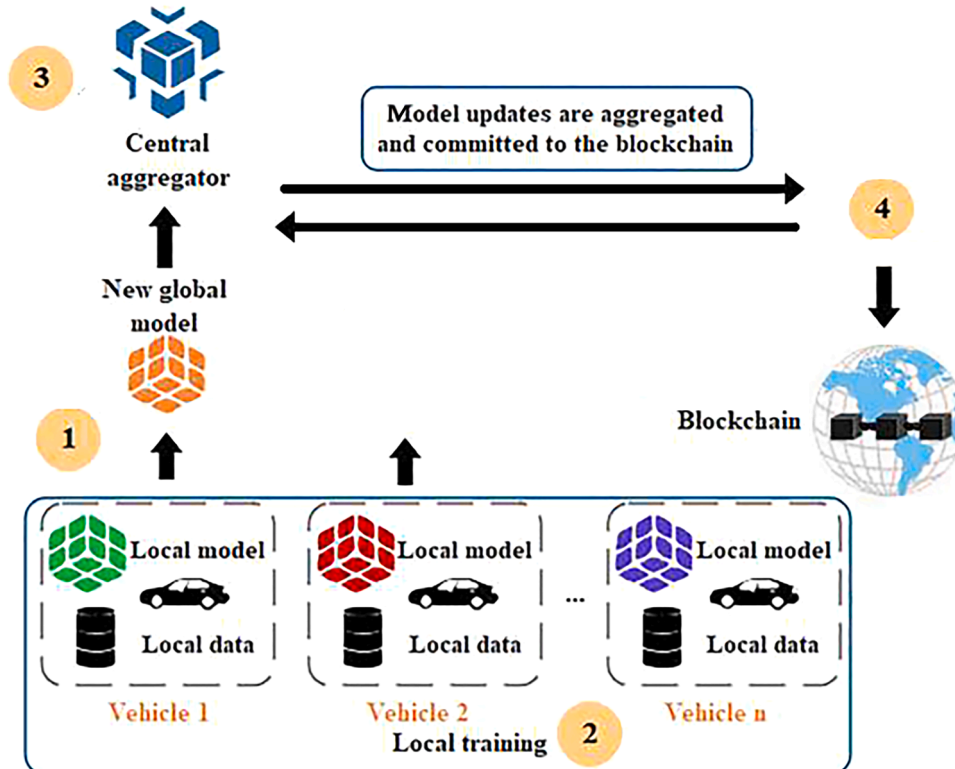


Fig. 3. Proposed architecture.

validation loss for early stopping, and optimizing memory usage in the GRU cells. By leveraging these optimization techniques, the GRU model can be trained more efficiently, improve its generalization capabilities, and overcome common challenges associated with RNN-based models. However, it's important to note that the effectiveness of these techniques may vary depending on the specific dataset and problem, and thorough experimentation and evaluation are necessary to determine the optimal configuration for achieving the desired performance of the GRU model.

The MANN (Memory Augmented Neural Network) model specified consists of two layers. The first layer is an LSTM (Long Short-Term Memory) layer with 16 units, which is capable of capturing long-term dependencies and sequential patterns in the data. The second layer is a Dense layer with six units and a softmax activation function, responsible for producing the final output probabilities. The model complexity of the MANN is determined by the number of layers, the number of LSTM units, and the connections between them. To optimize the MANN model, various techniques can be employed. These include careful weight initialization methods, appropriate activation function selection, regularization techniques like L1 or L2 regularization to prevent overfitting, dropout to improve generalization, gradient clipping to address exploding gradients, learning rate scheduling strategies, monitoring validation loss for early stopping, and memory management within the LSTM layer. Additionally, MANNs often incorporate memory modules or attention mechanisms to enhance their ability to store and retrieve information, and optimizing these components can also contribute to improved performance. Optimization of the MANN model requires iterative experimentation and evaluation, as the effectiveness of different techniques may vary depending on the specific dataset and problem at hand. By applying these optimization techniques, the MANN model can be trained more efficiently, improve its generalization capabilities, and overcome challenges associated with memory-based architectures.

The CW-RNN (Continuous Waveform Recurrent Neural Network) model specified consists of three layers. The first layer is a GRU (Gated Recurrent Unit) layer with 16 units and the "return_sequences=True" parameter, indicating that it returns the full sequence of outputs rather than just the final output. The second layer is another GRU layer with 16 units, and the third layer is a Dense layer with six units and a softmax activation function, responsible for generating the final output probabilities. The model complexity of CW-RNN is influenced by the number of layers, the number of units in each GRU layer, and the connections between them. GRU layers are a type of recurrent neural network (RNN) that can effectively capture sequential dependencies in the data. By returning the full sequence of outputs in the first GRU layer, the model can retain information from all time steps, which can be beneficial for certain tasks. To optimize the CW-RNN model, various techniques can be applied. These include appropriate weight initialization methods, activation function selection, regularization techniques such as L1 or L2 regularization to prevent overfitting, dropout to improve generalization, gradient clipping to address exploding gradients, learning rate scheduling strategies, monitoring validation loss for early stopping, and memory management within the GRU layers. Additionally, exploring different hyperparameters and architectures specific to CW-RNN, such as the number of units in each layer and the time step size, can also contribute to model optimization. Optimizing the CW-RNN model requires careful experimentation and evaluation, as the effectiveness of different techniques may vary depending on the specific dataset and problem being addressed. By applying these optimization techniques, the CW-RNN model can be trained more efficiently, enhance its generalization capabilities, and effectively capture dependencies in continuous waveform data.

All models in this context use the same activation function, which is appropriate for the multiclass detection task at hand. This activation function maps input data to output probabilities, with values ranging from 0 to 5, representing the number of outputs (attacks) labeled (0, 1, 2,

3, 4, and 5) (Table 6).

The proposed model includes the aggregation model Federated averaging model (FedAVG) to detect the attacks in VANET using the VeReMi dataset. The FedAVG algorithm process is shown in the following architecture (See Fig. 3) the server collects locally trained models from each client and obtains a global model through weighted averaging using fixed aggregation weights (Fig. 4).

This algorithm used this predefined function:

$$W = \sum_k \alpha_k W_k \quad (1)$$

Where:

W : the global model

k : the client

α_k the fixed aggregation weights α from data at the clients.

W_k : the current model of client k .

The FedAVG passed by three crucial steps:

1. The weight scaling factor: determines the proportion of a client's local training data compared to the overall training data across all clients. To calculate this factor, we first determine the client's batch size, and then obtain the total size of the global training data. By dividing the client's batch size by the global training data size, we obtain the scaling factor.
2. The scale model weights: adjust the weights of each local model based on the scaling factor calculated in step (1).
3. The sum scaled weights: adds up all the scaled weights from each client.

Different algorithms, including MLP, LSTM, GRU, MANN, and CW-RNN, have been used as training algorithms in federated learning. The training process is shown in the Fig. 5 begins with 10 as the number of clients (vehicles) conducting independent training sessions using stochastic gradient descent (SGD) on their respective datasets. Following the training phase, both clients proceed to update their models. These updated models are then transmitted to a central server for aggregation.

On the server, the received models from the clients are combined through an averaging mechanism, leading to the enhancement of the global model. Subsequently, this improved global model is communicated back to the clients. In a synchronized manner, the clients replace their previous models with this updated global model. This entire sequence of steps is iterated multiple times, typically around 100 rounds, to progressively refine the global model. Through this iterative approach, various clients can collaboratively contribute to the enhancement of the global model's performance, all the while maintaining the decentralization of their data. Before the test phase, the global model was saved and serialized to HDF5 to be used by the blockchain mechanism.

The SGD standard model was also used in our approach, which follows a centralized training approach for intrusion detection in VANET. With the SGD model, a single model is trained on the VeReMi dataset, enabling us to analyze the network and identify potential attacks. We used the same parameters as in the FL model, with 100 number of rounds (epochs), ensuring a fair comparison between the two approaches. However, it's important to note that the SGD model operates on a centralized server, which requires transferring the entire dataset and

Table 6
Model parameters choices.

Parameters	Models
SoftMax	Activation function
Categorical_Crossentropy	Loss function
SGD	Optimizer
100	Epochs
16	Batch size
0.01	Learning rate

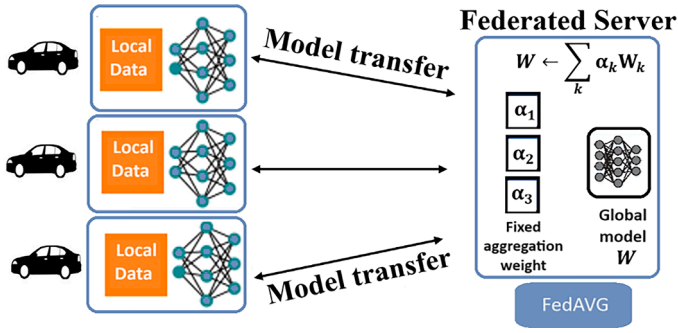


Fig. 4. FedAVG architecture.

Number of clients (vehicles) = 10

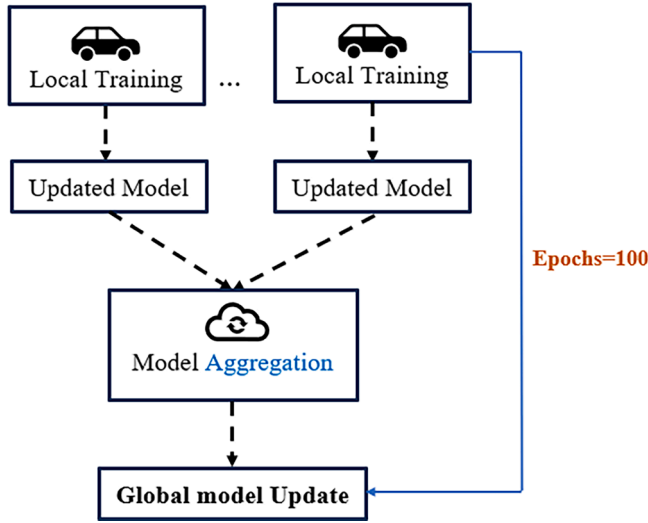


Fig. 5. FedAVG training.

raises privacy concerns.

The SGD model follows a centralized training approach on a central server, while the FedAVG model employs federated learning with multiple client devices. The SGD model uses a centralized dataset, whereas the FedAVG model ensures data privacy by keeping data local to the client devices and sharing only model updates. Privacy and security concerns differ between the models, as the SGD model requires access to the entire dataset, potentially raising privacy issues. The FedAVG model addresses this by keeping data decentralized, and preserving individual client privacy.

Communication efficiency also differs, with the SGD model involving communication between the central server and clients for data transfer, while the FedAVG model optimizes communication by aggregating model updates instead of transferring raw data. By understanding these differences, one can make informed decisions when choosing between the SGD and FedAVG models for intrusion detection in VANET, considering factors such as training approach, data security and privacy, communication efficiency, and the distributed nature of the VANET environment.

The process of integrating federated learning and blockchain, as illustrated in Fig. 5, encompasses several significant stages. To begin, vehicles contribute by developing local models based on their unique data characteristics. These models encapsulate insights specific to each vehicle's dataset. Federated learning showcases collective intelligence as these local models combine to form an enhanced Global Model. This collaborative process leverages the individual wisdom of each vehicle while maintaining data privacy.

The Global Model is then saved in Hierarchical Data Format Version 5 (HDF5) before being converted into a suitable format, often Hexadecimal, to ensure compatibility with the blockchain. This format is chosen to seamlessly integrate into the blockchain structure, ensuring data consistency and security during the transition.

Integration into the blockchain is orchestrated by a smart contract through its "set" function. Smart contracts, self-executing scripts, process the incoming Global Model data and embed it in the blockchain's immutable ledger. Hash functions are used to generate cryptographic fingerprints, validating both the Global Model's data integrity and the blockchain's block structure.

Hash functions add a layer of security, safeguarding data authenticity and preventing unauthorized modifications. By computing and comparing hash values, any discrepancies can be swiftly detected, preserving the data's unaltered state. This process elegantly combines federated learning, blockchain, and security measures to create a robust framework for data integration and protection.

3.2.4. Scalability and effectiveness of the proposed framework

This study proposes a framework that combines federated learning (FL) and blockchain for intrusion detection in Vehicular Ad hoc Networks (VANETs). The research demonstrates the scalability of the framework in terms of blockchain overhead and the effectiveness of federated learning with a large number of participating vehicles.

By leveraging FL, the framework distributes model training among vehicles, ensuring privacy preservation. The proposed approach utilizes various neural network models for intrusion detection. The research evaluates the framework's performance using quantitative evaluation metrics. The results show that the Federated Trained Model outperforms a traditionally trained Stochastic Gradient Descent (SGD) model in terms of precision and recall across various attack types. The FL-based model achieves higher precision, ranging from 68 % to 94 %, and consistently higher recall, ranging from 57 % to 88 %. These results highlight the superiority of FL in detecting intrusions, leading to gains in accuracy, recall, and precision.

Furthermore, the integration of blockchain technology enhances security and privacy protection in the collaborative FL training process. The use of blockchain ensures data integrity and facilitates trustworthy anomaly monitoring across decentralized nodes. Overall, the research demonstrates the scalability of the proposed framework, showcasing its effectiveness in intrusion detection across potentially thousands of participating vehicles. The combination of FL and blockchain addresses VANET vulnerabilities, enabling privacy-preserving and collaborative abnormality recognition. The results support the wider adoption of vehicular technologies and emphasize the importance of combining FL and blockchain for maintaining reliability, safety, and trust in VANET operations.

Our proposed solution is designed to work reliably and effectively in varied and unexpected network conditions. VANETs are subject to constant changes, including fluctuations in transmission rates, connectivity between vehicles, and signal disruptions. To address this variability, our solution utilizes adaptive intrusion detection mechanisms based on learning models. These models can dynamically adjust to adapt to the evolving network conditions in real-time. Moreover, our solution is designed to be resilient to network failures. In VANET environments, communication outages or temporary failures of certain vehicles are common. To ensure continuous intrusion detection, our solution can detect and isolate faulty vehicles while maintaining surveillance for operational vehicles.

We have also considered sophisticated attacks that VANETs may encounter. Our solution incorporates advanced detection mechanisms capable of recognizing and responding to common attacks such as denial-of-service attacks or false information attacks. Furthermore, it is designed to be scalable, enabling it to adapt to emerging attack techniques.

Regarding privacy and security, our solution emphasizes the

protection of sensitive user data. By employing federated learning, the detection models are trained locally on participating vehicles without the need to share individual data. This ensures that user information remains confidential while providing an effective intrusion detection mechanism.

In summary, our solution offers a flexible and adaptive approach for intrusion detection in VANETs, capable of operating reliably in varied and unexpected network conditions. It can adjust to network fluctuations, handle failures, detect sophisticated attacks, and safeguard user privacy. Thus, it represents a robust solution for VANET security, even in complex and changing situations.

3.2.5. Time complexity of proposed framework

In this section, we provide a guarantee on the time complexity of the proposed algorithm. In a neural network, the number of entities in each feature map is at most a constant multiplied by the number of input entities, denoted as n (typically, the constant is < 1). The convolution operation, which applies a fixed-size filter over an input signal with n features, takes $O(n)$ time. This is because each output is computed as the sum of products between certain features, denoted as k , in the input and a fixed number of weights, denoted as w , in the filter. Importantly, both w and k remain constant regardless of the value of n . Similarly, any pooling operation, whether max-pooling or average-pooling, also takes no more than linear time with respect to the input size. Moreover, the peripheral nodes can be computed in parallel, ensuring that the overall execution time remains linear, i.e., $O(n)$.

4. Experimental results

The application of smart contract consists of two main parts the smart contracts realization and the integration of blockchain in the IDS system. The realization of smart contracts is illustrated in the workflow presented in Fig. 6. The first step consists of creating etaMask, which is defined as a cryptocurrency wallet and browser extension for interacting with Ethereum-based applications. It allows users to manage their Ethereum accounts, store digital assets, and securely sign transactions directly from their web browsers. The second step details the development of the smart contract using Solidity in the Remix IDE. The contract will handle storage operations and involve MetaMask for ETH transactions. After this step, we declare the contract and define a variable in which we will store the data. The next step includes the implementation

of both functions get and set to stock the data and retrieve it. After completing the development of the smart contract, we will compile it in the Remix IDE. Before the last step, we deploy our contract on the MetaMask environment using the Injected provider. After deployment, we receive the transaction result, which includes the address of our account, the hash function for data decoding, and the gas price paid to deploy the contract. Finally, we execute the transformations, so We add and execute our deferred model, obtaining hashed data through the set function. Then, we use the get function to retrieve the model without altering any variables.

4.1. Integration of blockchain in the IDS system

The integration of the blockchain in the IDS system is shown in the workflow illustrated in Fig. 7. In our smart contract, we declare the Solidity language version, define variables and their types, and

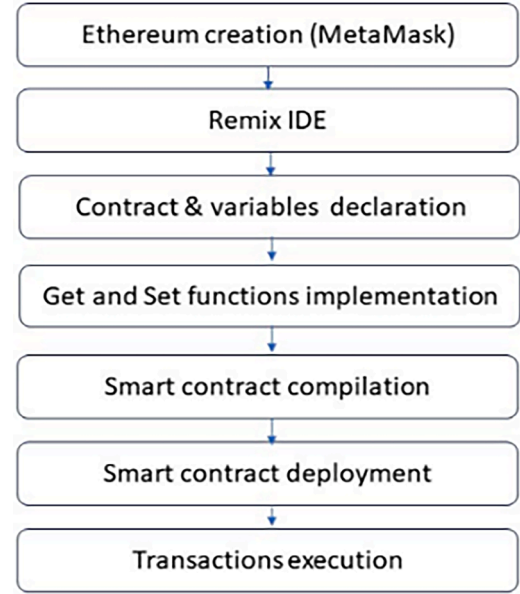


Fig. 7. Smart contract workflow.

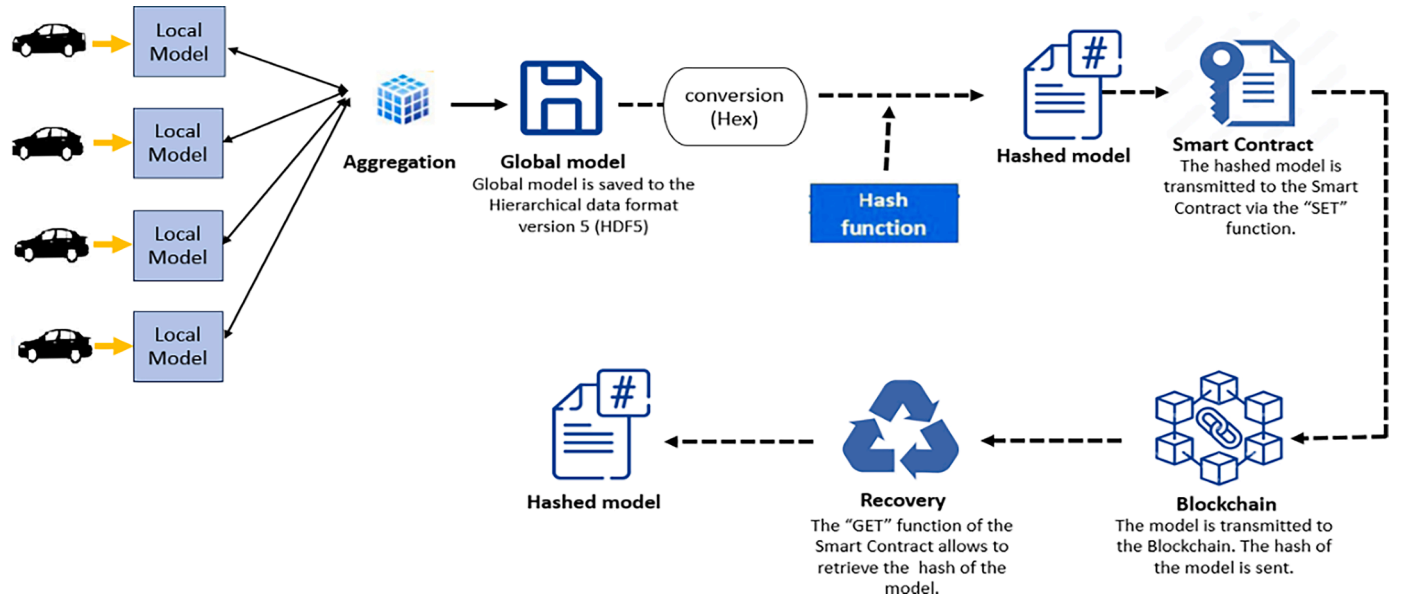


Fig. 6. FL with the blockchain network.

implement the function to send IDS system models to the Blockchain and retrieve them. After obtaining Ethereum, we deploy the contract using Metamask. Later, we will require the ABI contract and the contract address.

We established a connection to an Ethereum node, opting for Infura as the provider. Infura enables quick connection to a remote Ethereum node by specifying its endpoint. The final step involved calling the smart contract using the contract address and the ABI script, which grants access to the contract's functionalities. We invoked the "mem.functions.set(message)" function to interact with the contract, utilizing the "build-Transaction" method to create a transaction. Lastly, we confirmed the transaction payment directly using the Private Key from our MetaMask account (Fig. 8).

The integration of blockchain into the federated model offers in our approach enhanced security by leveraging the immutability and transparency provided by blockchain technology. This integration addresses the challenges in securing federated learning, ensuring data integrity, and establishing a distributed consensus mechanism. By incorporating blockchain, we can mitigate malicious activities and intrusions, strengthen trust in data sharing, and provide a robust framework for privacy preservation. This approach holds significant promise to maintain the security and privacy of collaborative learning environments while advancing the field of federated learning.

4.2. Results and discussion

In this section, we discuss the experimental results obtained from our study regarding the federated model and the SGD standard model. In addition, we compared our proposed approach with other previous works.

4.2.1. Classification metrics

Evaluation metrics are essential for achieving optimal classification performance during training. In our study, we used accuracy, recall, and precision metrics to evaluate intrusion detection using the VeReMi dataset. The VeReMi dataset covers a wide range of traffic scenarios, including highways, cities, and streets, making it suitable for capturing the dynamic environment of VANETs. In the current research, the VeReMi dataset is utilized in conjunction with federated learning and SGD approaches to classify misbehavior related to five different types of attacks. VANET environments are highly dynamic by nature, and it is crucial to assess their operational performance accurately. By incorporating the VeReMi dataset, this research ensures that the evaluations consider the dynamic aspects of VANETs. The dataset's inclusion of diverse traffic scenarios enables a comprehensive evaluation of misbehavior detection models under various driving conditions, vehicle densities, and road layouts. This allows for a more realistic assessment of the

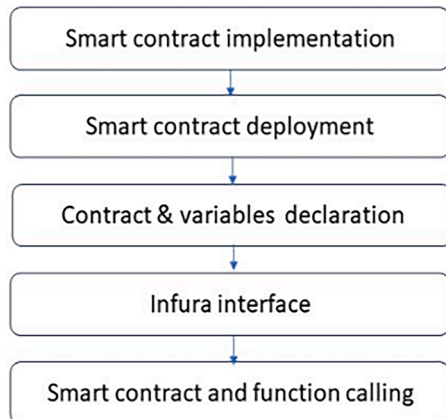


Fig. 8. Blockchain in IDS system.

model's effectiveness in dynamic VANET environments, accounting for rapidly changing network topologies, intermittent connectivity, and varying traffic patterns. By leveraging the VeReMi dataset, this research provides insights into the performance of federated learning and SGD approaches in detecting and addressing misbehavior across different attack types in the dynamic context of VANETs.

Table 7 presents the classification metrics of the federated model using the five models and Table 8 presents the classification metrics using the SGD standard model.

According to Figs. 9, and 10, we can make the comparison between the federated model and the SGD model. Based on the experimental results, it can be observed that the performance of the models varied when trained using both the FL and SGD approaches. The accuracy metric measures the overall correctness of the predictions, while the recall metric focuses on the ability of the models to correctly identify instances of positive class (attacks) among all the relevant instances. Among the models, MLP, LSTM, GRU, and MANN achieved relatively similar accuracy and recall scores, ranging from 80 % to 82 %, regardless of whether FL or SGD was used for training. This indicates that these models were able to learn the underlying patterns in the data and make accurate predictions. However, the CW-RNN model showed a significant difference in performance between the FL and SGD approaches. When trained using FL, it achieved an accuracy and recall score of 82 % as shown in Fig. 11, comparable to the other models. However, when trained using SGD, the accuracy dropped to 72 %. This suggests that the CW-RNN model may be more sensitive to the training approach and could benefit from the collaborative nature of FL. Overall, these results demonstrate that different models may perform differently when trained using FL or SGD. The CW-RNN model, in particular, showed a notable improvement in accuracy and recall when trained using FL compared to SGD. This highlights the potential of FL as an effective approach for training complex models in the context of intrusion detection in VANET. Regarding the precision metric, the CW-RNN model achieved the highest precision score of 77 % in the Federated Learning setting, outperforming the MLP, LSTM, GRU, and MANN models. This suggests that the CW-RNN model was effective in accurately predicting the target variable or class labels. However, its precision score dropped significantly to 52 % when trained using the SGD optimization algorithm. This implies that the CW-RNN model's performance suffered in the SGD setting compared to FL. Further investigation is needed to understand the reasons behind this discrepancy and potentially enhance the model's performance with SGD.

In conclusion, the experimental results indicate that the Federated Learning model outperformed the standard SGD model in all metrics tested. The CW-RNN model achieved the highest scores across the metrics, highlighting the superior performance of the FL approach, which enables decentralized training on local data. However, it is essential to consider additional factors such as scalability, resource requirements, and potential limitations of FL before drawing a definitive conclusion.

We have analyzed the precision and recall value achieved for each attack using two training approaches in given Table 9. The result shows that the Federated Trained Model consistently outperforms the SGD Trained Model in both Precision and Recall metrics for most attacks, indicating its superior ability to accurately identify attacks and minimize false positives and false negatives.

Table 7
Federated model classification metrics.

	Accuracy	Recall	Precision
MLP	80 %	80 %	74 %
LSTM	81 %	81 %	75 %
GRU	81 %	81 %	75 %
MANN	81 %	81 %	75 %
CW-RNN	82 %	82 %	77 %

Table 8
SGD model classification metrics.

	Accuracy	Recall	Precision
MLP	82 %	82 %	77 %
LSTM	82 %	82 %	77 %
GRU	81 %	81 %	76 %
MANN	82 %	82 %	77 %
CW-RNN	72 %	72 %	72 %

As the severity of the attacks increases from Attack1 to Attack16, both models face challenges, but the Federated Trained Model exhibits a more consistent and overall better performance across all attack types. Specifically, the Federated Trained Model excels in Precision for Attack1, Attack8, and Attack16, while the SGD Trained Model marginally outperforms it in Precision for Attack2 and Attack4. Similarly, the Federated Trained Model performs exceptionally well in Recall for Attack1, Attack2, Attack4, and Attack16, while the SGD Trained Model shows slightly higher Recall for Attack8. In summary, the Federated Trained Model proves to be the more reliable and effective option, demonstrating higher Precision and Recall values across various attack scenarios compared to the SGD Trained Model. The results obtained by our approach are also compared with recent related work for intrusion detection in terms of precision, as illustrated in Table 10 and Fig. 11.

The results demonstrate that our approach performs competitively or even outperforms recent work in most attack scenarios. It shows higher effectiveness in defending against Attack2 and Attack4, suggesting that our approach may provide better protection against certain types of security threats.

The proposed approach is also compared with several recent state-of-the-art methods for intrusion detection in terms of accuracy in Table 11.

Fig. 11 shows the performance comparison of the recent methods with the proposed work over the VeReMi dataset.

4.2.2. Integration of blockchain with FL model

To verify the proper functioning of our contract, we can validate the contract connection and read the hash data. This verification can be observed in Fig. 12, where the model hash is displayed using the print function. Moving forward, with the aid of Infura, we can examine the Blocks and Transactions associated with the IDS models and the Blockchain. These elements were mentioned earlier and can only be decrypted through the utilization of the recovery function implemented within the smart contract.

To conduct further verification of our application's functionality, we employed the retrieval function 'get'. Fig. 12 illustrates the syntax of

this function along with the displayed result. The purpose of this function is to retrieve the same models that were initially sent to the Blockchain. The utilization of the recovery function is implemented within the smart contract.

The integration of homomorphic encryption (HE) with the federated learning (FL) model was evaluated through experimental analysis. The results of this integration are presented in the following Table 12, which showcases the accuracy scores achieved by each implemented model.

We can make comparison observations between the federated and the federated encrypted approaches. From these results, we can see that in most cases, the models trained with federated learning and encryption achieved comparable or slightly lower accuracy compared to the models trained in the regular federated learning setting without encryption. However, the differences in accuracy are relatively small, indicating that the use of encryption during federated learning did not significantly degrade model performance (Fig. 13).

Using encryption during federated learning is essential for maintaining data privacy and security, especially when data is distributed across multiple devices or servers. Even though there might be a slight trade-off in model accuracy, the benefits of protecting sensitive data and ensuring privacy outweigh the small reduction in accuracy.

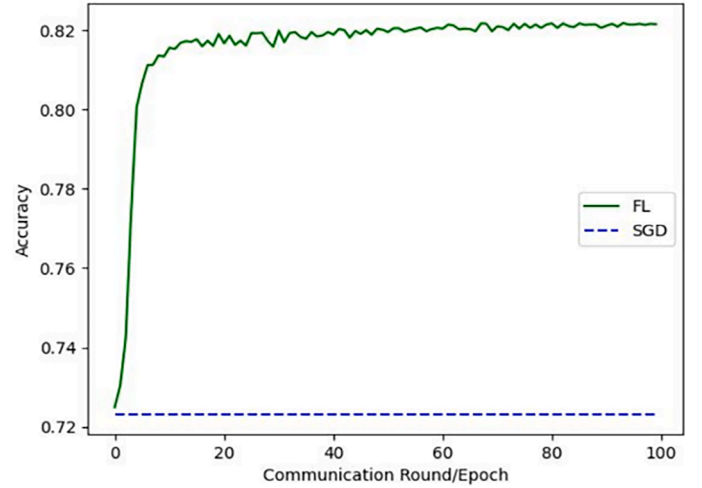


Fig. 10. FL vs SGD accuracy.

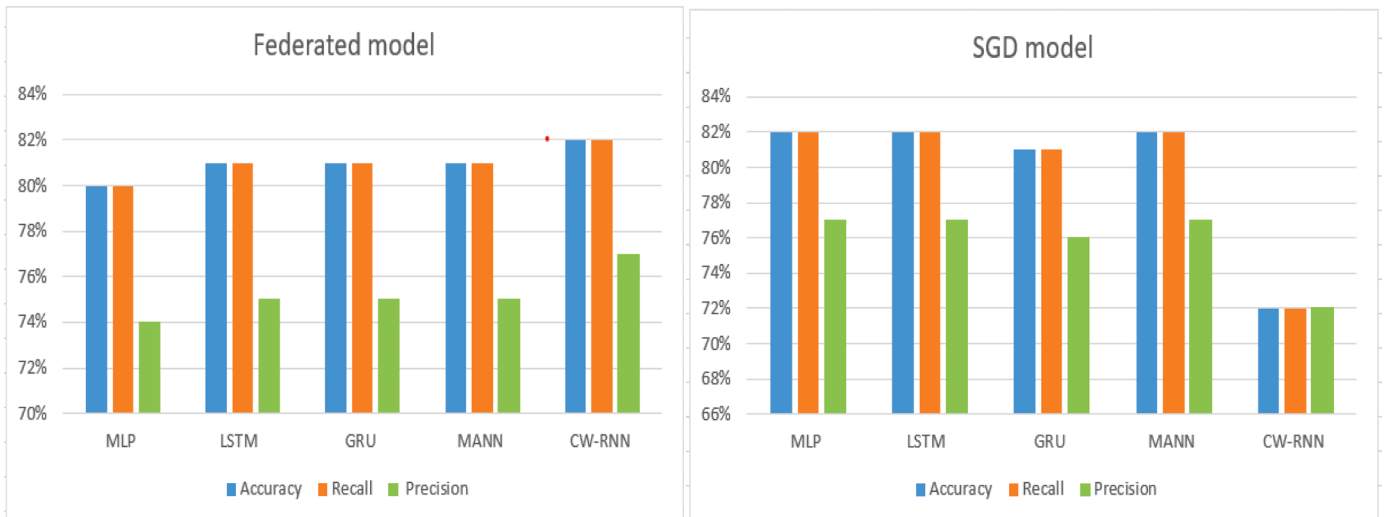


Fig. 9. FL & SGD classification metrics.

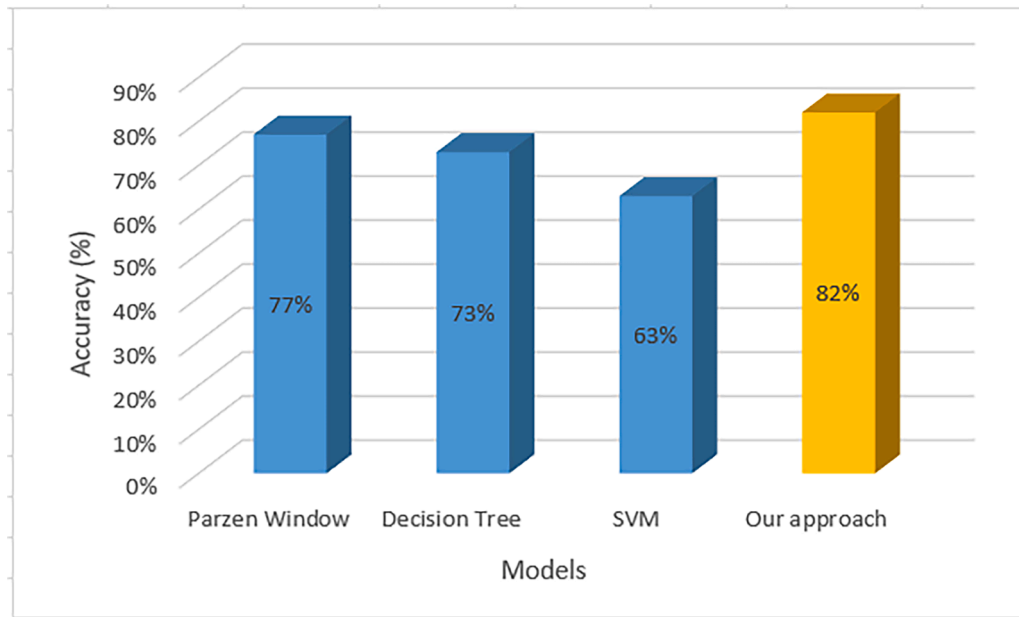


Fig. 11. Comparison of our approach with recent methods in terms of accuracy.

Table 9

Federated model classification metrics.

Precision	Attack1	Attack2	Attack4	Attack8	Attack16
Federated Trained Model	94 %	68 %	70 %	87 %	92 %
SGD Trained Model	83 %	62 %	24 %	53 %	78 %
Recall					
Federated Trained Model	88 %	57 %	61 %	63 %	80 %
SGD Trained Model	83 %	48 %	48 %	56 %	73 %

Table 10

Our approach vs recent work.

	Our approach	Recent work [42]
Attack1	94 %	94 %
Attack2	68 %	65 %
Attack4	70 %	46 %
Attack8	87 %	86 %
Attack16	92 %	92 %

Table 11

Comparison with recent methods.

Model	Accuracy
ML-MDS [43]	80 %
Parzen Window [44]	77 %
Decision Tree [45]	73 %
SVM [45]	63 %
Our approach	82 %

4.2.3. Discussion of results

The method NNFLB-NET for intrusion detection in VANETs, which integrates federated learning (FL) and blockchain (BC), yielded promising experimental results. Collaborative learning through FL significantly improved the accuracy of intrusion detection models, with CW-RNN achieving an impressive accuracy score of 82 % in the federated learning approach, surpassing the traditional SGD model. The integration of blockchain technology enhanced the security and trustworthiness of the FL process, offering an additional layer of protection against

adversarial attacks.

Every proposed solution has its advantages and also has gaps. We have succeeded in implementing an intelligent approach for identifying intrusions in VANETs by integrating federated learning and blockchain and federated learning with HE. However, we consider our current solution to be a naive one. We have the potential to enhance it further by incorporating additional models and exploring cloud computing implementation within the integrated technologies. By doing so, we can refine and optimize our solution, ensuring its effectiveness and robustness in detecting intrusions in VANETs. In the context of enhancing security in VANETs, both blockchain technology and homomorphic encryption serve the common goal of improving data protection and privacy. As part of our future work, we aim to investigate and compare the differences between these two security approaches in the context of VANETs, with a specific focus on utilizing federated learning. By studying the distinct characteristics and benefits of blockchain and homomorphic encryption within the federated learning framework, we seek to determine which approach offers the most effective and efficient security solution for VANETs, taking into account factors such as data privacy, scalability, and performance. Other future work will focus on integrating additional communication and security protocols. By incorporating protocols such as DSRC or C-V2X, our goal is to explore how our FL and Blockchain-based approach can seamlessly interface with real-world VANET environments. We aim to ensure a smooth integration of our approach, enabling effective communication and interaction within VANET networks.

5. Conclusion

In summary, our research focused on integrating federated learning (FL) and blockchain technology to detect intrusions in VANET networks, utilizing the VeReMI dataset. We implemented various neural network models, including MLP, LSTM, GRU, MANN, and CW-RNN, following the FL approach. Additionally, we compared these models with the standard SGD approach.

Our results unequivocally demonstrated the superiority of FL over the SGD model for intrusion detection. Collaborative learning through FL yielded higher accuracy, recall, and precision metrics, highlighting its effectiveness in the VANET environment. FL proved to be a powerful technique for enhancing security, and privacy in VANET networks.

```

C:\Users\ASUS\Downloads\memoire\venv\Scripts\python.exe C:/Users/ASUS/Desktop/blockchaincode.py
model hash : 3da1d7c8eab0256bc6344ec057c27eafc3cc23be
contact is connected True
hash function : b'y\xd6\xbe\xd7\xa4\x9e\xf2\xa2\xb4\xe0\x1c\x8c\xa7\x8b\xbd0H\xa4\xd4\xaa\xde\x9c\xbe\xd2\x9e*.J\n7'
3da1d7c8eab0256bc6344ec057c27eafc3cc23be
Process finished with exit code 0

```

Fig. 12. Contract operating result.

Table 12
Encrypted federated model accuracy.

Models	Accuracy_Enc_Fed	Accuracy_Fed
MLP	69 %	80 %
LSTM	70 %	81 %
GRU	72.94 %	81 %
MANN	72.31 %	81 %
CW-RNN	72.31 %	82 %

Furthermore, we successfully integrated our FL model with blockchain technology, leveraging smart contract technology. This integration effectively addressed challenges related to securing FL, ensuring data integrity, and preserving privacy. Our results showed that the integration of FL with blockchain further improved intrusion detection, providing an additional layer of security and trust in the VANET ecosystem. In conclusion, our study synthesizes the benefits of combining FL and blockchain for intrusion detection in VANETs. We have validated the superiority of FL over the traditional SGD model and effectively tackled security challenges through blockchain integration. Our research contributes to enhancing the overall security posture of VANET networks, highlighting the significance of an integrated approach. Moving forward, our findings open avenues for further advancements in securing intelligent transportation systems and ensuring the safety and privacy of VANET networks. By integrating FL and blockchain, we offer a robust framework that addresses the evolving challenges of intrusion detection in VANETs, paving the way for a more secure and resilient vehicular communication ecosystem.

The experimental results presented in this paper have substantiated

the efficacy of our proposed method. However, several challenges and avenues for further research remain to enhance intrusion detection in VANETs and ensure optimal security and privacy. We have made progress by incorporating cloud computing into our approach, although this aspect is still in the developmental phase. We have successfully implemented an intelligent approach for intrusion identification in VANETs, leveraging the VeReMi dataset and integrating federated learning and blockchain technologies. Moving forward, our future work will focus on refining our solution through the implementation of cloud computing techniques, aiming to enhance its overall effectiveness, and investigating and comparing the differences between both blockchain and homomorphic encryption approaches using federated learning in the context of VANETs. As another perspective of our work, we will thoroughly examine the considerations and challenges associated with deploying the proposed framework in real-world vehicular ad hoc network (VANET) scenarios, placing particular emphasis on hardware requirements, compatibility with existing vehicular communication systems, deployment challenges, security, and privacy, as well as validation and evaluation. We recognize the significance of these real-world deployment considerations and strive to provide practical and feasible solutions to ensure a successful integration of our framework in real-world VANET scenarios.

Proposing additional comparisons between our proposed framework and other state-of-the-art intrusion detection systems in VANETs, beyond SGD models, may be an interesting perspective to this work. This will be our future work, integrating different comparison techniques and incorporating real-world VANET scenarios. It is our goal to continue exploring and improving our approach to better address the complexities of intrusion detection in VANETs. By integrating cloud computing,

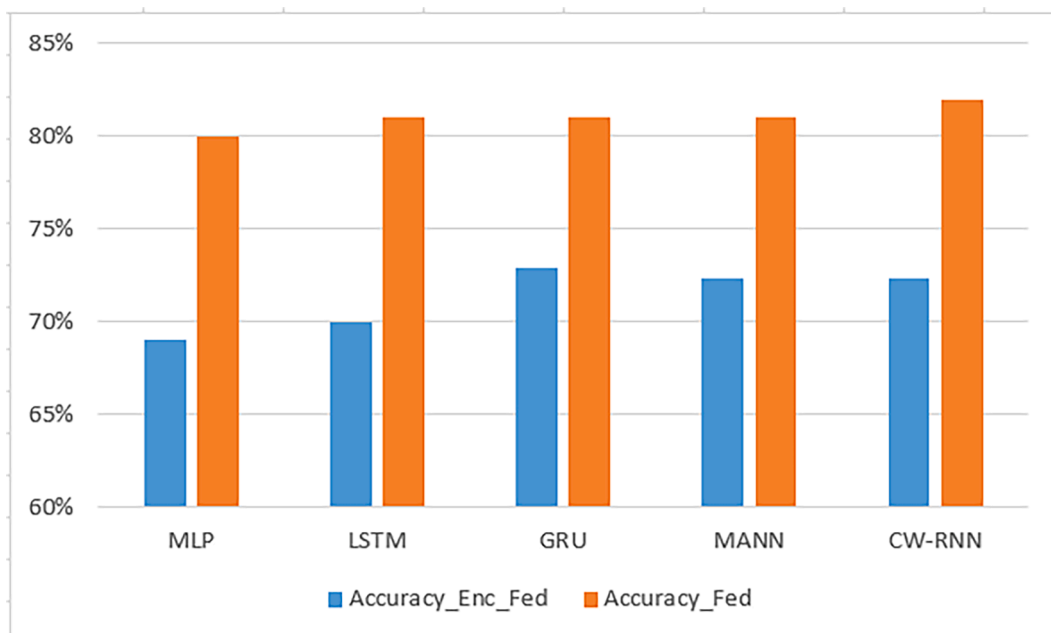


Fig. 13. Accuracy encrypted federated vs accuracy federated.

we anticipate further advancements in our solution and its potential to strengthen the security and privacy of VANET networks.

CRedit authorship contribution statement

Fedwa Mansouri: Writing – original draft, Validation, Conceptualization. **Mounira Tarhouni:** Writing – original draft, Validation, Project administration, Methodology. **Bechir Alaya:** Visualization, Supervision, Project administration, Formal analysis. **Salah Zidi:** Writing – review & editing, Visualization, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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